

ARYAN MONDAL

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EDUCATION

University of Massachusetts Amherst

MS in Computer Science (Incoming)

Amherst, Massachusetts

Sep 2026 – May 2028

SRM Institute of Science and Technology

B.Tech in Computer Science And Engineering, CGPA: 8.39/10

Chennai, Tamil Nadu

Sep 2022 – May 2026

- **Relevant Coursework:** Cryptography and Network Security, Forensics and Incident Response, Security Management, Computer Networks, Cyberwarfare, Mobile and Wireless Security, Hacker Techniques, Tools, and Incident Handling

EXPERIENCE

Security Intern

Thirumoolar IT Solutions

Feb 2025 – Mar 2025

Chennai, Tamil Nadu

- Implemented 20+ vulnerability scans and penetration tests using OWASP ZAP, Burp Suite, and Nmap to establish a missing security baseline, identifying 40+ issues and achieving a 25% improvement in overall security posture.
- Investigated codebases of 3 internal web applications where unreviewed code posed active exploit risk, applying manual analysis to surface SQL Injection and XSS vulnerabilities and enabling developers to resolve a significant share of security flaws.
- Researched and summarized emerging threats in response to a rapidly evolving landscape, synthesizing intelligence reports to strengthen the internal knowledge base and increase team response readiness.

PROJECTS

Network Intrusion Detection System

Apr 2025 – Apr 2026

- Engineered a hybrid NIDS to address prohibitive compute costs on edge devices, combining a 4-layer DNN (~68K params) with a pseudo-GNN branch (~8K params) via adaptive late fusion to achieve 99.64% accuracy on 60,000 CIC-IDS-2017 flows at 1,250× fewer parameters than Transformer-based models.
- Designed a learnable sigmoid-normalized fusion weight α to eliminate manual ensemble tuning, which autonomously converged from 68.9% DNN to 61.1% GNN over 25 epochs, empirically establishing that relational features outperform statistical ones for intrusion detection.
- Optimized an end-to-end raw-socket-to-classification pipeline on PyTorch to meet real-time edge deployment demands, achieving 16,000+ flows per second at under 5 ms latency with a false positive rate of 0.08%.

PUBLICATIONS

An Adaptive Feedback Architecture for Stabilizing SNSPDs in TF-QKD Networks using SDQN

Under Review

IEEE Transactions on Quantum Engineering (TQE)

- Engineered an SDQN with hot-swappable PID and Bayesian controllers to counter SNSPD performance collapse under real-world thermal stress, achieving a 41% improvement in Secret Key Rate and a 4× reduction in SKR variance (σ_{SKR} : 0.180 $\rightarrow$ 0.038).
- Modeled photon detection efficiency via Ginzburg–Landau critical current equations to address PID overshoot caused by the nonlinear bias–temperature relationship, deploying a Matérn 5/2 Gaussian Process with Expected Improvement to reach 95% of the optimum in 80 cycles versus 120 for PID alone.
- Implemented Mahalanobis Distance anomaly detection across correlated SKR, QBER, temperature, and bias metrics to catch physical-layer attacks that passive countermeasures miss, triggering an automated Security Mode clamp upon a 3σ breach to defeat blinding and thermal manipulation.

A Robust and Lightweight Authentication Protocol for UAV Swarm Communications using HKDF-based Key Derivation and AEAD

Jul 2026

IEEE CONECCT 2026

- Designed R-HAAP to replace a cryptographically weak Duffing chaotic-map keystream with a verified ECDH $\rightarrow$ HKDF $\rightarrow$ ChaCha20-Poly1305 pipeline, achieving 150–250 μ s per-packet latency on ARM Cortex-M4 — 20–33× faster than the original protocol and immune to cache-timing side-channel attacks.

- Implemented per-packet key derivation to eliminate replay and forgery risk from reused session keys, constructing a unique HKDF.expand info string from UAV ID and monotonic offset counter to cryptographically bind metadata to the Poly1305 MAC under dual replay countermeasures.
- Validated swarm scalability to address unproven crypto overhead at scale, simulating star and FANET mesh topologies across 10–100 UAVs in UavNetSim-v1 and achieving packet delivery rate above 99.8% and end-to-end latency under 15 ms.

Lightweight Hybrid DNN-GNN Architecture for Network Intrusion Detection with Adaptive Late Fusion

Apr 2026

ICNWC 2026

- Designed a hybrid NIDS to retain relational traffic learning without expensive graph construction, integrating a 4-layer DNN (~68K params) with a pseudo-GNN MLP (~8K params) via adaptive late fusion to achieve 99.64% accuracy on 9,000 held-out test samples with 80K total parameters.
- Analyzed a learnable scalar α to quantify statistical versus relational discriminative power, finding it shifted from 68.9% DNN to 61.1% GNN over 25 epochs, with ablation confirming the 8K pseudo-GNN matched the full hybrid while outperforming the 68K DNN-only baseline by 0.12 percentage points.
- Benchmarked the model against ConvNeXt-Tiny (28M params) and a Transformer NIDS (100M params) to establish a new efficiency frontier, achieving competitive or superior accuracy at $350\times$ and $1,250\times$ fewer parameters with a false positive rate of 0.08%.

Quantum Key Distribution for SCADA in Power Grids: Security Architectures and Practical Applications

Nov 2025

ICETCE 2025

- Developed SVM and Random Forest anomaly detection classifiers for QKD-enhanced SCADA where conventional cryptography is insufficient against quantum threats, training on 10,000 simulated data points with linear-kernel SVM achieving 94.90% accuracy and perfect anomaly recall (1.00).
- Evaluated four classifiers — SVM, Random Forest, KNN, and Neural Network — under class imbalance to identify the optimal model for critical-infrastructure deployment, comparing precision (0.68–0.70), recall (0.94–1.00), and F1-score (0.80–0.81) and demonstrating KNN’s complete failure on the anomaly class.
- Established a replicable QKD-SCADA security framework to address the lack of standardized quantum-resilient grid architectures, researching 30+ field deployments and mapping BB84, decoy-state, and SDN-monitored QKD protocols to SCADA threat vectors including FDIA, MitM, and ransomware.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Bash, SQL

Cyber Security & Networking: Cryptography, Digital Forensics, Anomaly Detection, Malware Analysis, Penetration Testing, Web Exploits (SQLi, XSS), Operating Systems, SIEM/SOC Operations, SCADA Security, UAV Security, Quantum Key Distribution (QKD)

Tools & Frameworks: Claude Code, OWASP ZAP, Burp Suite, Nmap, Wireshark, Scapy, PyShark, Apache Kafka, FastAPI, Grafana, TensorFlow, Keras

Other Skills: Risk Management, Compliance & Regulation, Security Awareness, Threat Intelligence, Strategic Planning, Project Management